Reg. No.															

B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2025

Fifth Semester

21CSC301T - FORMAL LANGUAGE AND AUTOMATA

(For the candidates admitted from the academic year 2021-2022 to 2024-2025)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours Max. Marks: 75

PART - A (20 x 1 = 20 Marks) Marks BL CO PO

Answer **ALL** Questions

1. Which of the following pair of regular expression are not equivalent? 1 1 1 1

   A) 1(01)* and (10)*1 B) x(xx)* and (xx)*x

   C) (ab)* and a*b* D) x+ and x*x+

2. The minimum number of states required in a DFA (along with dead state) to check whether the 3rd bit is 1 or not for |n| >=3 over {0,1} 1 1 1 1

   A) 3 B) 4

   C) 5 D) 1

3. Reverse of a DFA can be formed by 1 1 1 1

   A) making final state as non-final B) making final as starting state and starting state as final state

   C) using PDA D) None of the mentioned

4. Expressive power of NFA & ε-NFA is 1 1 1 1

   A) Different B) Same

   C) ε-NFA is more powerful D) NFA is more powerful

5. The maximum number of steps that machine M takes on any input of size N is _____ 1 1 5 1

   A) Time Complexity B) Space Complexity

   C) Time Space Complexity D) None

6. If the number of steps required to solve a problem is O(N^k) , then the problem is said to be solved in 1 1 5 3

   A) Non-polynomial time B) Polynomial time

   C) Infinite time D) None of the mentioned

7. Which type of symbols contain in the stack of PDA 1 1 3 2

   A) Variable B) Terminal

   C) Epsilon D) Variable and Terminal

8. S→ AB/CD, A→BG/0, B→AD/ε, C→ CD/1, D→ BB/E 1 1 2 3
   E→ AF/B1, F→EG/0C, G→AG/BD
   Identify the true statement from the list below.

   A) E is not nullable B) G is not nullable

   C) C is not nullable D) B is not nullable

9. The symbol Zo in PDA definition denote 1 1 3 3

A) Initial state B) Initial stack alphabet

C) Variable D) Stack content

10. Push down machine represents 1 1 3 3

A) Type 3 grammar. B) Type 2 grammar

C) Type 1 grammar D) Type 0 grammar

11. Any Language generated by an unrestricted grammar is: 1 1 4 3

A) Recursive B) Context free Language

C) Regular Language D) Context sensitive

12. Given the grammar G: 1 1 2 3

1. S-->AS

2. S-->AAS

3. S-->SA

4. A-->aa

Which of the production denies format of CNF?

A) 2,4 B) 1,3

C) 1, 2, 3, 4 D) 2, 3, 4

13. A string is accepted by a PDA when 1 1 3 2

A) Stack is empty B) Acceptance state

C) Both (A) and (B) D) Rejected state

14. Given a Grammar G:S->aA,A->a,A->B,B->A, B->bb. Which among the following will be the simplified grammar? 1 1 2 3

A) S->aA, A->a | bb, B-> a | bb B) S->aA , A->a, B->bb

C) S->aA , A->a, B->A | bb D) S->aA, A->B, B->A, B->b

15. Consider the following grammar 1 1 2 3

S-->0A | 1B

A-->0AA | 1S | 1

B-->1BB | 0S | 0

Which of the following is TRUE?

A) Grammar is not ambiguous B) Grammar is ambiguous

C) Grammar does not generate 001110 D) Both B and C

16. Recursive Enumerable Languages are closed under 1 1 4 2

A) Union only B) Intersection only

C) Both Union and Intersection D) Neither Union nor Intersection

17. The problems which have no algorithm, regardless of whether or not they are accepted by a turing machine that fails to halts on some input are referred as: 1 1 5 3

A) Decidable B) Undecidable

C) Computable D) Finite

18. Turing machine uses 1 1 4 2

A) Binary representation B) Ternary representation

C) Unary representation D) Normal representation

19. Which of the following problem is not NP-hard? 1 1 5 3

A) Hamiltonian circuit problem B) The 0/1 Knapsack problem
C) Finding Bi-connected component of a graph D) The graph coloring

20. If $\delta(q, a) = (p, y, R)$, then 1 1 4 2

A) qabxcd |-- ypbxcd B) qabxcd |-- ybpxcd
C) qabxcd |-- apybxcd D) qabxcd |-- pybxcd

PART - B (5 x 8 = 40 Marks) **Marks BL CO PO**

Answer **ALL** the Questions

21 a. 1. Illustastrate the formal description of DFA and NFA with a suitable examples. 8 2 1 1
2. Construct a deterministic finite automata which accepts all set of strings which never contains a substring "xxy" for all set of strings formed using $\Sigma=\{x,y\}$.

(OR)

b. 1. Compare and contrast between Mealey and Moore machine? Also construct a Moore Machine that computes divisibility of binary number by 3. 8 2 1 1
2. Can we design a regular expression for a paliondrome? Justify your answer.

22 a. Remove all unit-productions, all useless productions, and all null-productions from the grammar: 8 0 2 3

G: S-->aA | aBB
A-->aaA | ε
B--> bB | bbC
C-->B.

What language does this grammar generate?

(OR)

b. Convert the given context free grammar into its equivalent grebaich normal form(GNF) 8 3 2 3

S-->AA | a

A-->SS | b

23 a. A deterministic pushdown automaton (DPDA) and a non-deterministic pushdown automaton (NPDA) accepts which set of languages? Design a pushdown automata for a language $L=\{a^ib^jc^k : i = j \text{ OR } j=k \text{ for } \Sigma=\{a,b,c\}$. 8 3 3 2

(OR)

b. Construct a context free grammar generating the following language and hence equivalent PDA accepting the language. 8 3 3 2

$L=\{a^nb^ma^n : m,n>=1\} \cup \{a^nc^n : n>=1\}$

24 a. Give the tuple definition of Turing Machine(TM) with instantaneous description. Construct a TM for language $L=\{www ; w \in (a,b)^*\}$. 8 2 4 2

(OR)

b. Write short notes on ANY two of the following: 8 3 4 2
1. Extensions of TM
2. Universal Turing Machine
3. Recursive and Recursively Enumerable Languages

25 a. 1. Does the PCP with two lists A = (y, yxy^3 ,yx) and B = (y^3 , yx, x) have a solution? If yes give the solution. 8 3 5 3

2. Discuss NP-hard and NP-complete problems with suitable examples.

(OR)

b. Compare and contrast Decidable and Undecidable Languages with examples? 8 3 5 3

PART - C (1 x 15 = 15 Marks) **Marks BL CO PO**

Answer **ANY ONE** Questions

26 1. State and Prove the pumping lemma for context free languages Using the pumping lemma find out whether the language L={$a^n b^n c^n$: n>=1} is context free or not. 15 3 2 3

2. Minimize the given FA

1.

Q/Σ	Σ=a	Σ=b
-->q0	q1	q2
***q1**	q3	q4
***q2**	q4	q3
q3	q5	q5
q4	q5	q5
***q5**	q5	q5

27 Design a Turing machine which computes multiplication of two numbers. 15 4 4 2

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